Capstone Project: Design an AI Solution

AI Solution Architect Program | Friday 16 October 2026 | Instructor: Nagarjuna Naidu

What you will do

On Days 4 and 5 your team builds the Northwind Knowledge Assistant in five milestones (assistant and tools, MCP, tracking and monitoring, security and explainability, demo and design). That working assistant is your reference implementation. For the capstone, choose one scenario below (or your own domain), adapt the design to it with the canvas, demo what you built and present your architecture and trade-offs. A clear, defensible architecture matters more than extra code.

Your building tool: the Architecture Canvas

You have been filling the canvas section by section during the week. Use it as the backbone of your design and presentation.

Choose a scenario

Scenario	What it is	Constraints	KPIs to consider
A SME loan assistant (bank)	A relationship-manager copilot that answers lending-policy questions from internal documents with citations, and screens loan applications with a default-risk model that can explain its output.	Regulated lending (explainable decisions, audit trail, assume customer data must stay in-country); p95 latency under 3 seconds; several thousand documents that change monthly.	Time to answer, share of answers with correct citations, risk-model quality in business terms (cost of wrong approvals and rejections), override rate.
B Claims triage and customer assistant (insurer)	Triage incoming claims (forms, photos, notes), summarise them for adjusters, flag possible fraud, and answer customer status questions.	Personal and possibly health data; human review for every adverse decision; peak load after weather events; multiple legacy systems.	Cycle time per claim, fraud caught vs false alarms, adjuster hours saved, customer satisfaction.
C Support and retention assistant (retail or telecom)	A support assistant that answers from product and policy documents, looks up orders and accounts through tools, and triggers retention offers using a churn model.	Seasonal traffic peaks, strict cost per conversation, CRM and ERP integration, offers must follow policy.	Resolution rate, cost per contact, churn reduction, escalation rate.
D Your own domain	Healthcare, manufacturing, public sector, energy, education or another domain you know. Write 3 lines: problem, data, main constraint.	Agree the scenario with the instructor in the first 10 minutes.	Define 3 KPIs.

Deliverables (one set per team)

- A one-page architecture diagram (boxes and arrows: data, retrieval or models, tools, serving, integration, governance).
- The completed Architecture Canvas (9 sections).
- An evaluation and KPI plan: how you will know it works before and after launch.
- A risk list with mitigations (security, bias, compliance, cost, latency).
- A rough cost and latency estimate, and your MVP scope.
- A short live demo of your assistant (about 2 minutes of your 8).

Schedule

Time	Activity
Milestone 5	Demo your assistant; adapt the design to your scenario with the canvas; draw the architecture; prepare 5 slides at most
8 min per team	Present (1 min problem, 3 min architecture, 2 min trade-offs, 1 min risks, 1 min questions)
15 min	Feedback and discussion

How you will be scored (100 points)

Criterion	Points
Fit to the problem and KPIs	20
Architecture quality and justified choices (rules, ML, RAG, agent, cloud)	25
Data, retrieval and governance	15
Evaluation and operations (testing, monitoring, release)	15
Responsible AI, security and compliance	15
Clarity and defence of trade-offs	10

Ground rules

- Use synthetic or public information only. Do not use real client data.
- You may use any free cloud or open-source tools; name services in plain terms and note that names change.
- The demo uses your milestone notebooks; keep it short.
- Every choice needs a reason: say what you rejected and why.